

ALIBABA CLOUD AI HACKATHON PAKISTAN 2026 · HEALTHCARE

BloodBridge AI

Emergency blood coordination for Pakistan —
every drop, matched in seconds. Safely.

Safety-first donor matching

Self-verifying demand forecasts

Privacy by default

Team BloodBridge — working product, seeded with 6 cities across Pakistan

Finding blood in an emergency means hours of desperate phone calls.

1 Matching is slow and manual

Families call relatives, post on Facebook, and drive from bank to bank — while the patient's clock runs out. There is no single place where a verified need meets willing, **eligible** donors.

2 Safety rules get ignored

Donors are contacted even when they are medically ineligible to donate again. Pakistan's guideline requires a 90-day rest for men and 120 for women — informal networks don't check.

3 Blood banks can't see demand coming

Inventory decisions are made on instinct. Rare groups run out without warning — while donor phone numbers circulate publicly and get misused.

Who lives this problem

Patients & families — emergencies, childbirth, surgery, accidents.

Thalassemia & dialysis patients — Pakistan carries one of the world's largest thalassemia burdens; many need a transfusion every 2–4 weeks, for life.

Hospitals & blood banks — coordinating under time pressure with no forecasting.

Voluntary donors — willing to help, but contacted chaotically and without eligibility checks.

One platform, three coordinated systems.

1 Safety-first donor matching

Ranks compatible donors by blood type, distance, donation recency and reliability — and **hard-blocks anyone inside the mandatory resting window.**

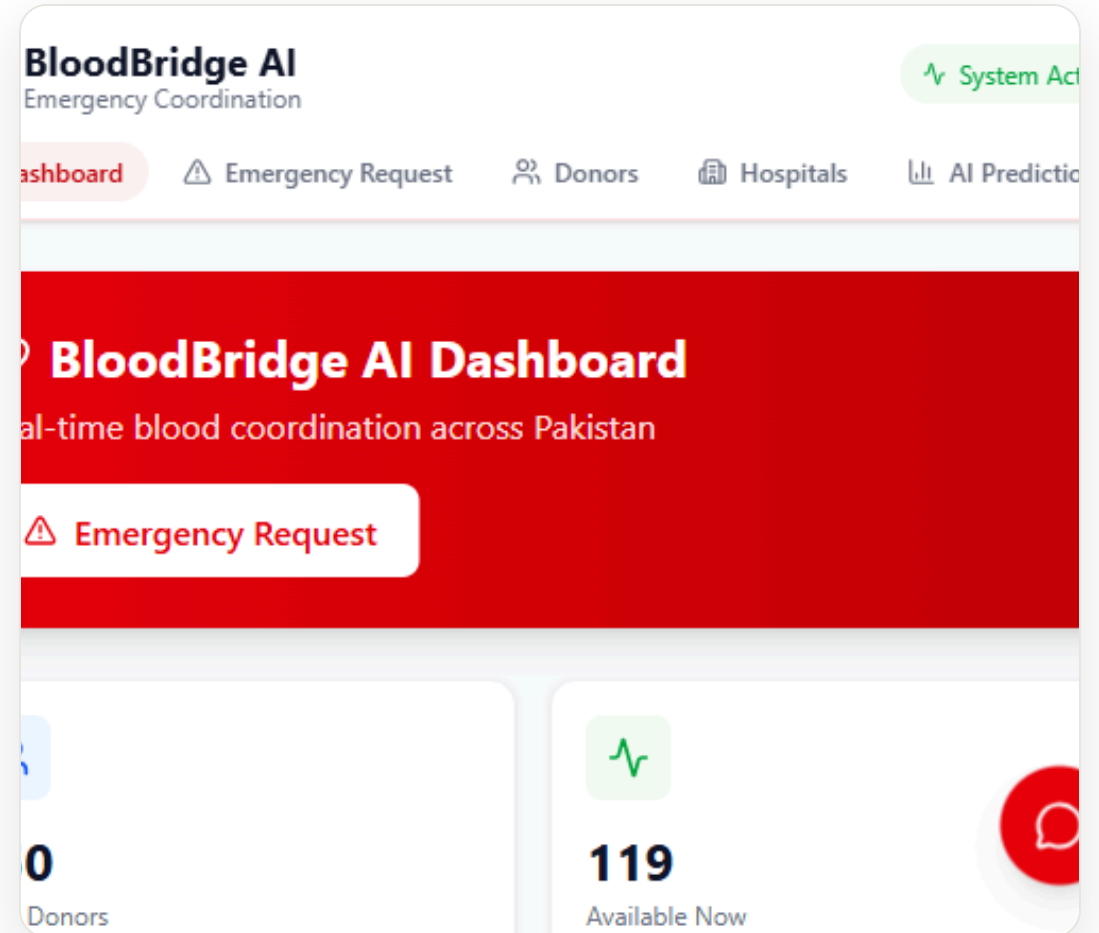
2 Self-verifying demand forecasts

7-day predictions per city & blood group — with **backtested accuracy displayed live in the product**, so the numbers are proven, not claimed.

3 Assistant chatbot

Instant answers on compatibility, eligibility rules, live inventory and how to submit a request.

Audience: hospital & blood-bank staff post verified requests · voluntary donors register and respond · coordinators forecast and plan stock.



Live dashboard — inventory, donors by city, requests (patient names masked)

THE NEED · THE IMPACT IT MAKES

From a phone tree to a **match in seconds** — with safety enforced by design.

10

ranked, medically eligible donors surfaced instantly per request — ranked by compatibility, distance & ETA

17

donors automatically blocked in one live demo request for being inside the 90/120-day resting window

7 days

of demand forecast per city & blood group, with shortage warnings before banks run dry

0

patient names or donor phone numbers exposed in public views — masked by default

Why it matters: every emergency that starts with a match instead of a phone tree saves the most expensive currency in medicine — time. And every blocked ineligible donor is a complication that never happens. The system also gets **smarter with use:** real requests and fulfillments feed the forecast model.

Matching that **refuses** to recommend ineligible donors.

 **BloodBridge AI**
Emergency Coordination

System Active

[Dashboard](#) [Emergency Request](#) [Donors](#) [Hospitals](#) [AI Predictions](#)

[Back to Dashboard](#)

 **AI Donor Match Results**

Patient: **Muhammad Bilal** | Hospital: **Shaukat Khanum Memorial Hospital**

B+
Blood Group

2
Units

Critical
Urgency

Matched
Status


10
Top Matches


10
Compatible Donors



Weighted compatibility score

Blood-type compatibility (40%) · haversine distance (30%) · donation recency (20%) · donor rating (10%). Critical requests re-weight toward distance.

The hard safety rule

Donors inside the 90-day (men) / 120-day (women) resting window are **excluded entirely** — never just ranked lower. The count is shown live on every match ("Blocked by Resting Window").

Closing the loop

When staff confirm which donors actually donated, those donors automatically enter their resting window — the system keeps itself medically honest.

Predictions that **prove their own accuracy.**



The model

Weighted 14-day moving average + linear-regression trend + weekday seasonality, per city and blood group — explainable by design.

Backtested, not just claimed

The model is re-run on past days using only prior history, then compared with what actually happened — **~80–84% accuracy across 112 forecasts**, displayed live in the product.

Learns from reality

Every real request and fulfillment updates demand history — the forecast improves as the system is used, instead of living on seed data.

A data loop that keeps the system truthful — and privacy by default.



Privacy by default

- ✓ Patient names masked everywhere public: “M*** B***” — contact numbers never returned
- ✓ Donor phones masked in the directory: 0335..... — real numbers revealed only through the AI match for an active request
- ✓ All inputs validated at the API boundary — malformed or malicious data is rejected with clear errors

Trust model (mobile prototype)

- ✓ Only **verified hospital / blood-bank staff** can post requests — institutions submit registration numbers
- ✓ Donors complete an **8-point health self-screening** (re-screened in person at the bank, as required)
- ✓ Patient identity replaced by a private internal reference on the shared feed

THE TECHNOLOGY

Simple stack, engineered honestly.

Architecture

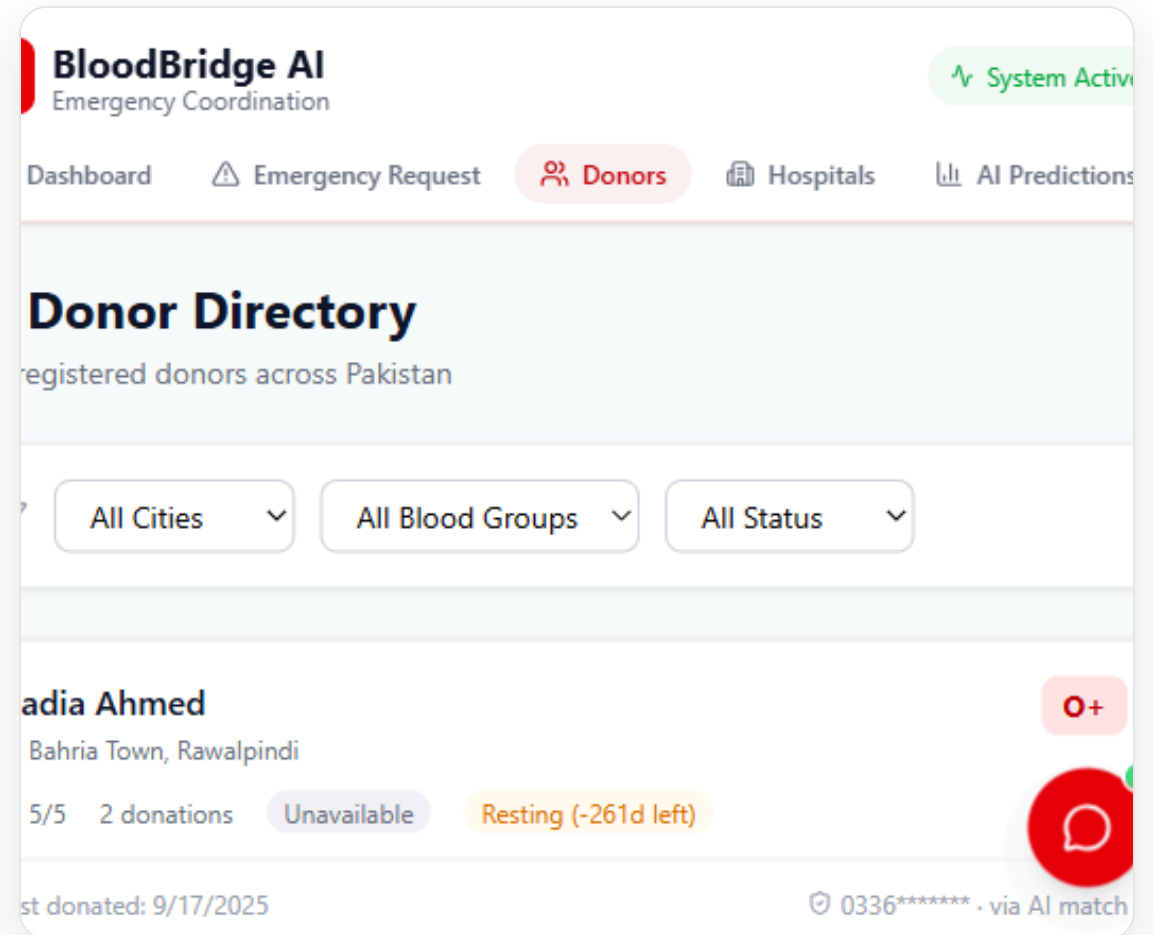
React 19 + Vite + Tailwind + Recharts single-page app → **Node.js / Express** API → in-memory store with **automatic disk persistence** (survives restarts; Mongoose-ready schemas for a MongoDB upgrade).

Quality engineering

21-test automated API suite covering the full lifecycle (request → match → fulfill → backtest) — all passing. Input validation, masking and eligibility live on the **server side**, so the rules can't be bypassed from the browser.

Realistic dataset

150 donors, 19 real hospitals & blood banks, 90 days of demand history across Lahore, Karachi, Islamabad, Rawalpindi, Faisalabad and Multan.



Donor directory — masked phones, live eligibility status per donor

Everything in this deck **runs today**, from the repo.

- ✓ **Full working app** — 6 pages, 10+ API endpoints, chatbot, live matching & forecasting; two commands to run locally (**`npm run install:all`**, **`npm run dev`**)
- ✓ **Complete request lifecycle** — create → AI match → call donors → confirm fulfillment → donors rest, bank restocks, model learns
- ✓ **Tested end-to-end** — 21 automated API tests, all passing, included in the repo
- ✓ **Second working prototype** — the mobile trust model (role onboarding, institution verification, donor screening) runs standalone in any browser
- ✓ **Honest scope** — prototype persistence is JSON-on-disk; production needs auth, SMS alerts and hospital integration. We say so in the README — and the roadmap is concrete.

Judges can verify in 2 minutes

1. Clone the repo, run two commands
2. Submit a request in the UI
3. Watch the AI match — with ineligible donors blocked
4. Check Analytics — accuracy is backtested, live
5. Run **`node server/smoke-test.js`** — 21 green checks

ROADMAP & CLOSE

**Next: SMS/WhatsApp alerts, map view,
blood-unit expiry tracking, MongoDB + auth.**

The coordinates, eligibility data and forecast loop are already in place — the roadmap builds on what runs today, not on promises.

Every drop, matched in seconds. Safely.